

Evaluating Expressions with Radicals

MA.8.NSO.1.7

I can evaluate an expression that contains square roots, cube roots and integer exponents.

Yesterday every number was a measurement.

Today the radical returns — its bar is a grouping symbol.

Warm-Up

On your own. Four minutes. No notes.

1. How many significant digits are in 0.0250 ?
2. $(3 \times 10^4)(2 \times 10^{-7})$
3. Find the value: $\sqrt{196}$
4. Evaluate: $20 - 3 \cdot 4$

Warm-Up

Answers.

1. How many significant digits are in 0.0250 ? **3**
2. $(3 \times 10^4)(2 \times 10^{-7})$ **6×10^{-3}**
3. Find the value: $\sqrt{196}$ **14**
4. Evaluate: $20 - 3 \cdot 4$ **8**

Notes

Copy all three lines.

I. The radical bar is a grouping symbol

$$\sqrt{25} - \sqrt{36} + \sqrt{1} = 5 - 6 + 1 = 0$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4$$

$$\sqrt{9 + 16} = \sqrt{25} = 5, \quad \text{never} \quad 3 + 4$$

Everything under the bar is evaluated before the root is taken.

Notes

II. Square roots and cube roots are not the same shape

$$\sqrt{49} = 7 \qquad \sqrt[3]{64} = 4 \qquad \sqrt[3]{-64} = -4$$

$$\sqrt{\frac{1}{9}} = \frac{1}{3} \qquad \sqrt[3]{\frac{1}{125}} = \frac{1}{5}$$

An odd root keeps the sign of its radicand. A square root sign never gives a negative value.

Notes

III. Radicals alongside exponents

$$5^{-3} = \frac{1}{5^3} = \frac{1}{125} \qquad \sqrt{4^3 - 39} = \sqrt{25} = 5$$

$$\sqrt{4^3 - 39} \cdot 5^{-3} = 5 \cdot \frac{1}{125} = \frac{1}{25}$$

Finish each piece — the radical, the power — and only then put them together.

Example 1

A power and a radical in one expression.

$$\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$$

Evaluate.

Example 1

Worked.

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4} \quad \text{square the whole base, sign included}$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4 \quad \text{finish under the bar, then take the root}$$

$$\frac{1}{4} + 4 = \frac{17}{4} \quad \text{combine}$$

Answer: 17/4

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

Your Turn

Answer.

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

1/9 minus 2

Answer: $-17/9$

Example 2

A radical inside a product.

$$\sqrt[3]{27} - 1.4(\sqrt{3^2 - 5})$$

Evaluate.

Example 2

Worked.

$$\sqrt[3]{27} = 3, \quad 3^2 - 5 = 4 \quad \text{the two innermost pieces}$$

$$\sqrt{4} = 2, \quad 1.4(2) = 2.8 \quad \text{the root, then the product}$$

$$3 - 2.8 = 0.2 \quad \text{subtract last}$$

Answer: 0.2

Your Turn

Same steps, your numbers. Boards up when done.

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

Your Turn

Answer.

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

−2 plus 10

Answer: 8

Whiteboards • Question 1 of 9

Boards up on three.

$$\sqrt{121}$$

Answer it.

What times itself?

Whiteboards • Question 1 of 9

Answer.

$$\sqrt{121}$$

$$11 \cdot 11$$

Answer: 11

Whiteboards • Question 2 of 9

Boards up on three.

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

$$\sqrt{169}$$

Answer: 13 feet

Whiteboards • Question 3 of 9

Boards up on three.

$$\sqrt[3]{-125}$$

Answer it.

What number, three times?

Whiteboards • Question 3 of 9

Answer.

$$\sqrt[3]{-125}$$

$$(-5)(-5)(-5)$$

Answer: -5

Whiteboards • Question 4 of 9

Boards up on three.

$$\sqrt{2^3 + 17}$$

Answer it.

Finish under the bar first.

Whiteboards • Question 4 of 9

Answer.

$$\sqrt{2^3 + 17}$$

$$8 + 17 = 25$$

Answer: 5

Whiteboards • Question 5 of 9

Boards up on three.

$$\sqrt{9 + 16}$$

Which is the value?

A. 7

B. 25

C. 12.5

D. 5

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$\sqrt{9 + 16}$$

Which is the value?

A. 7

B. 25

C. 12.5

D. 5

Answer: D — 5

Whiteboards • Question 6 of 9

Boards up on three.

$$\sqrt[3]{-64}$$

Which is the value?

A. -4

B. not a real number

C. -8

D. 4

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\sqrt[3]{-64}$$

Which is the value?

A. -4

B. not a real number

C. -8

D. 4

Answer: A — -4

Whiteboards • Question 7 of 9

Boards up on three.

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

Answer it.

Top and bottom each get the root.

Whiteboards • Question 7 of 9

Answer.

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

$\frac{1}{3}$ and $\frac{1}{3}$

Answer: $\frac{2}{3}$

Whiteboards • Question 8 of 9

Boards up on three.

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

Answer it.

Finish each piece, then multiply.

Whiteboards • Question 8 of 9

Answer.

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

5 times one hundred twenty-fifth

Answer: 1/25

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Simon says that $\sqrt[3]{-27}$ is not a real number,
because you cannot take the root of a negative.
Explain why he is wrong, and give the value.

Write your answer in sentences.

Which root is he thinking of?

Whiteboards • Question 9 of 9

Answer.

Simon says that $\sqrt[3]{-27}$ is not a real number,

because you cannot take the root of a negative.

Explain why he is wrong, and give the value.

a cube root is not a square root

Answer: -3

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. This skill continues tomorrow: one assignment, due at the start of the class after 4.06.

Today's skills — all required

Evaluate radical expressions — D8J